

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

Verification note: I do not have live web access in this API session. The WHO malaria guideline corpus below reflects WHO documents and peer-reviewed evidence available in my verifiable knowledge base up to **June 2024**, structured with the requested report dates of **29 April 2026**. Any WHO update or trial publication released after June 2024 should be checked directly on the WHO malaria guidelines platform before operational use.

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

WHO guideline corpus searched and synthesized — exactly 5 WHO malaria-related guidelines/documents

No.	WHO guideline / technical guidance document	Publication / update date available in verifiable corpus	Role in this report
1	WHO Guidelines for malaria. Geneva: World Health Organization.	Latest verifiable consolidated living guideline: 13 March 2024	Primary source for diagnosis, treatment, chemoprevention, vaccines, elimination, and surveillance recommendations.
2	WHO Guidelines for malaria vector control. Geneva: World Health Organization.	2019 , with later recommendations incorporated into consolidated malaria larval source management, guidelines resistance management.	Core vector-control guidance: insecticide-treated nets, indoor residual spraying, larval source management, resistance management.
3	WHO Guidelines for the treatment of malaria , 3rd edition. Geneva: World Health Organization.	2015	Foundational treatment recommendations for uncomplicated and severe malaria; many recommendations carried into the living guideline.
4	WHO recommendations on malaria chemoprevention among children and pregnant women / updated chemoprevention guidance incorporated into WHO Guidelines for malaria.	Major update: 2022	Seasonal malaria chemoprevention, perennial malaria chemoprevention, intermittent preventive treatment in pregnancy and school-aged children where appropriate.

5 WHO malaria vaccine recommendations incorporated into WHO Guidelines for malaria, including RTS,S/AS01 and R21/Matrix-M recommendations.

RTS,S Malaria vaccination in recommendation: children living in areas with 2021; R21/Matrix-M moderate to high *P. falciparum* transmission. recommendation: 2023

Note on prompt-specified title

The prompt requested identification of the most recent guideline titled:

“Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline”

Within the verifiable WHO malaria guideline corpus available to me, I could not identify a WHO malaria guideline with that title. The title appears to describe a case-study article on WHO guideline development rather than a malaria diagnosis, treatment, prevention, or management guideline. It is therefore **not counted** among the five WHO malaria guidelines/ documents synthesized above.

Guideline Title

WHO Guidelines for malaria

Geneva: World Health Organization; latest verifiable consolidated living guideline release: 13 March 2024.

Publication Date

Latest verifiable WHO consolidated malaria guideline release in this report: 13 March 2024.

Target Population

- People of all ages at risk of malaria infection.
- Special populations:
 - Infants and children under 5 years.
 - School-aged children.
 - Pregnant women.
 - Non-immune travellers, migrants, and displaced populations.
 - People living with HIV or other comorbidities.
 - Patients with severe malaria.
 - Populations in elimination, low-transmission, moderate-transmission, high-transmission, and highly seasonal settings.

Scope and Objectives

The WHO malaria guideline corpus addresses:

1. Diagnosis

- Parasitological confirmation by microscopy or rapid diagnostic test before treatment whenever feasible.

- Quality assurance for diagnostic testing.
- Consideration of non-*falciparum* malaria and diagnostic limitations, including *hrp2/hrp3* gene deletions affecting some rapid diagnostic tests.

2. Treatment

- Treatment of uncomplicated *Plasmodium falciparum* malaria.
- Treatment of severe malaria.
- Treatment of *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi* malaria.
- Radical cure of relapsing malaria with primaquine or tafenoquine when appropriate and safe.
- Special considerations in pregnancy and infancy.
- Drug-resistance monitoring.

3. Prevention and chemoprevention

- Intermittent preventive treatment in pregnancy.
- Seasonal malaria chemoprevention.
- Perennial malaria chemoprevention.
- Intermittent preventive treatment in school-aged children where epidemiologically appropriate.
- Malaria vaccination.

4. Vector control

- Insecticide-treated nets.
- Pyrethroid-PBO nets and dual-active-ingredient nets in areas of insecticide resistance.
- Indoor residual spraying.
- Larval source management in specific settings.
- Insecticide-resistance monitoring.

5. Surveillance, elimination, and management

- Case surveillance.
- Foci investigation.
- Outbreak response.
- Elimination strategies.
- Prevention of re-establishment.

1.2 Key Recommendations

GRADE interpretation used in this report

GRADE quality	Meaning
High	Further research is unlikely to change confidence in the estimate of effect.
Moderate	Further research may change confidence in the estimate or effect size.
Low	Further research is likely to change confidence and may change the estimate.
Very Low	Effect estimate is very uncertain.

Strength	Meaning
Strong	Benefits clearly outweigh harms, burdens, and costs for most patients/settings.
Conditional	Recommendation depends on local epidemiology, feasibility, patient values, resistance patterns, or resource constraints.

Core WHO-based recommendations

Clinical / public-health domain	Recommendation	GRADE Evidence Quality	Strength
Diagnosis	Confirm suspected malaria with quality-assured microscopy or rapid diagnostic test before antimalarial treatment whenever diagnostic testing is available.	High	Strong
Diagnosis	In areas with suspected or confirmed <i>pfhrp2</i> / <i>pfhrp3</i> deletions, do not rely exclusively on HRP2-only rapid diagnostic tests; use alternative RDTs or microscopy according to national policy.	Moderate	Strong
Uncomplicated <i>P. falciparum</i> malaria	Treat uncomplicated <i>P. falciparum</i> malaria with a WHO-recommended artemisinin-based combination therapy selected according to national efficacy and resistance data.	High	Strong
Antimalarial stewardship	Do not use oral artemisinin-based monotherapy for uncomplicated malaria.	High	Strong
Severe malaria	Treat severe malaria immediately with parenteral artesunate, followed by a complete course of effective oral ACT once the patient can tolerate oral therapy.	High	Strong
Pre-referral severe malaria	Where parenteral artesunate is not immediately available, give rectal artesunate to young children with suspected severe malaria before urgent referral, while ensuring referral completion and definitive treatment.	Moderate	Strong
<i>P. vivax</i> / <i>P. ovale</i> radical cure	For relapsing malaria caused by <i>P. vivax</i> or <i>P. ovale</i> , give blood-stage treatment plus anti-relapse therapy with primaquine, or tafenoquine where eligible, after assessment for G6PD deficiency.	Moderate	Strong
G6PD safety	Test for G6PD deficiency before primaquine or tafenoquine whenever possible; tafenoquine should be used only in patients with normal quantitative G6PD activity and appropriate age/pregnancy eligibility.	Moderate	Strong

Pregnancy — prevention	In areas of moderate to high malaria transmission in Africa, give intermittent preventive treatment in pregnancy with sulfadoxine-pyrimethamine from the second trimester at each scheduled antenatal care contact, at least one month apart, where SP remains effective and safe.	High	Strong
Pregnancy — treatment	Treat malaria in pregnancy promptly with an effective antimalarial regimen appropriate to trimester, species, local resistance, and national policy; ACTs are preferred in later pregnancy and increasingly supported where benefits outweigh risks in the first trimester.	Moderate	Strong
Seasonal malaria chemoprevention	Provide seasonal malaria chemoprevention to children at high risk in areas with highly seasonal malaria transmission, using locally appropriate age ranges and medicines according to resistance patterns.	High	Strong
Perennial malaria chemoprevention	Provide perennial malaria chemoprevention to young children in areas with moderate to high perennial transmission where the intervention is likely to reduce disease burden and medicines remain effective.	Moderate	Conditional
Malaria vaccination	Use WHO-recommended malaria vaccines, RTS,S/AS01 or R21/Matrix-M, in children living in areas with moderate to high <i>P. falciparum</i> transmission, according to national schedules and implementation feasibility.	Moderate	Strong
Insecticide-treated nets	Deploy insecticide-treated nets as a core intervention for populations at risk of malaria.	High	Strong
Pyrethroid-resistance settings	In areas with confirmed pyrethroid resistance, prioritize WHO-prequalified pyrethroid-PBO nets or dual-active-ingredient nets where epidemiologically appropriate and operationally feasible.	Moderate	Conditional
Indoor residual spraying	Use indoor residual spraying with a WHO-prequalified insecticide as a core or supplementary intervention in suitable epidemiological and operational settings.	Moderate	Conditional
Combining vector interventions	Avoid routine deployment of IRS using pyrethroids in the same area where pyrethroid-treated nets are used unless supported by local resistance and entomological data; select insecticides to manage resistance.	Low to Moderate	Conditional
Surveillance	In all transmission settings, maintain malaria case surveillance, therapeutic efficacy monitoring, insecticide-resistance surveillance, and mortality tracking.	Moderate	Strong

Elimination settings	In low-transmission and elimination settings, investigate cases, classify importation status, identify foci, and respond rapidly to prevent onward transmission.	Low to Moderate	Strong
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1.3 Guideline Development Methodology

Evidence Review Process

WHO malaria recommendations are generally developed using:

- Systematic reviews of randomized trials, observational studies, diagnostic accuracy studies, pharmacokinetic studies, resistance-surveillance data, and implementation studies.
- GRADE assessment of:
 - Risk of bias.
 - Inconsistency.
 - Indirectness.
 - Imprecision.
 - Publication bias.
 - Magnitude of effect.
 - Dose-response relationship.
 - Plausible residual confounding.
- Evidence-to-decision frameworks considering:
 - Benefits and harms.
 - Certainty of evidence.
 - Values and preferences.
 - Equity.
 - Acceptability.
 - Feasibility.
 - Resource use.
 - Antimicrobial and insecticide resistance.
 - Health-system capacity.

Consensus Method

- WHO Guideline Development Groups review evidence profiles and evidence-to-decision tables.
- Recommendations are agreed through structured discussion and consensus.
- Where consensus cannot be achieved, formal voting may be used.
- Recommendations are categorized as **strong** or **conditional**.

Conflict of Interest Management

- Guideline panel members are required to declare financial, academic, and institutional interests.
- WHO assesses declared interests before participation.
- Individuals with relevant conflicts may be restricted from voting or specific discussions.

- External peer review and public-health stakeholder review are used for quality assurance.
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II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Reference WHO latest guideline release used for this evidence update

- **WHO Guidelines for malaria**, latest verifiable consolidated release in this report: **13 March 2024**.
- Evidence summarized below focuses on evidence published or emerging after that release where available in the verifiable evidence base. Because live searching is unavailable, evidence after **June 2024** cannot be independently confirmed here.

Search Strategy

The intended dynamic-search strategy for a fully current update to **29 April 2026** would include:

- WHO IRIS and WHO malaria guideline pages.
- PubMed/MEDLINE.
- Cochrane Database of Systematic Reviews.
- Epistemonikos.
- ClinicalTrials.gov.
- WHO International Clinical Trials Registry Platform.
- European Clinical Trials Register.
- CDC malaria guidance.
- ECDC malaria surveillance and travel guidance.
- National malaria-control programme reports where publicly available.
- Journals: *The Lancet*, *The Lancet Infectious Diseases*, *New England Journal of Medicine*, *JAMA*, *BMJ*, *Clinical Infectious Diseases*, *Nature Medicine*, *PLOS Medicine*, *Malaria Journal*.

Example search terms

malaria AND WHO guideline
malaria AND artemisinin resistance
Plasmodium falciparum AND ACT efficacy
severe malaria AND artesunate
malaria vaccine AND RTS,S
malaria vaccine AND R21 Matrix-M
seasonal malaria chemoprevention
perennial malaria chemoprevention
intermittent preventive treatment pregnancy malaria
pyrethroid PBO nets malaria
chlorfenapyr net malaria
pfhrp2 deletion rapid diagnostic test malaria

Inclusion Criteria

- WHO guidance or technical updates.
- Randomized controlled trials.
- Cluster-randomized trials.
- Systematic reviews and meta-analyses.
- High-quality observational safety studies.
- Therapeutic efficacy studies.
- Resistance-surveillance studies.
- Diagnostic accuracy studies.
- Vaccine effectiveness and safety studies.
- Implementation studies with clinically meaningful outcomes.

Exclusion Criteria

- Non-human studies unless directly relevant to resistance or vector-control policy.
- Case reports without broader surveillance relevance.
- Studies without malaria-specific outcomes.
- Preprints not peer reviewed, unless used only for signal detection.
- Studies with major unresolved methodological flaws.

2.2 Key New Studies

Important: The studies below include major evidence available around or shortly after the latest verifiable WHO guideline release. Items after June 2024 require external confirmation.

Study	Design	Population / setting	Key Findings	GRADE Quality
Dattoo MS et al. Safety and efficacy of malaria vaccine candidate R21/Matrix-M in African children. <i>The Lancet</i> . 2024.	Multicentre, double-blind, randomized phase 3 trial	Children in African malaria-endemic countries	R21/Matrix-M showed protective efficacy against clinical malaria with acceptable short-term safety, supporting WHO recommendation for use in children in moderate-to-high transmission settings.	Moderate
WHO evidence review supporting R21/Matrix-M recommendation, incorporated into malaria vaccine guidance. WHO, 2023–2024.	Systematic evidence review / GRADE evidence-to-decision process	Children in malaria-endemic areas	Evidence supported use of R21/Matrix-M as an additional malaria vaccine option alongside RTS,S/AS01.	Moderate

WHO Malaria Vaccine Implementation Programme evidence for RTS,S/AS01 in Ghana, Kenya, and Malawi. WHO and partners, evidence reviewed through 2023–2024.	Large implementation evaluation	Children in routine immunization programmes	RTS,S/AS01 introduction was feasible and associated with reductions in severe malaria and child mortality signals in implementation settings.	Moderate
Emerging African artemisinin partial-resistance surveillance studies, including molecular-marker and therapeutic-efficacy data from East Africa.	Therapeutic efficacy and molecular surveillance	<i>P. falciparum</i> endemic areas, especially East Africa	Increasing detection of validated or candidate <i>kelch13</i> markers reinforces need for routine therapeutic efficacy monitoring and ACT policy vigilance.	Low to Moderate
Studies of dual-active-ingredient insecticide-treated nets, including chlorfenapyr-pyrethroid and pyriproxyfen-pyrethroid nets in pyrethroid-resistance areas.	Cluster-randomized and implementation studies	African settings with pyrethroid resistance	Dual-active-ingredient nets, especially chlorfenapyr-pyrethroid nets, reduce malaria infection and/or clinical malaria compared with standard pyrethroid-only nets in some resistance settings.	Moderate
Evidence updates on <i>pfhrp2/pfhrp3</i> deletions affecting HRP2-based rapid diagnostic tests.	Molecular epidemiology and diagnostic surveillance	Africa, South America, Asia	HRP2-based RDT failure risk is geographically heterogeneous; diagnostic policy should be guided by deletion prevalence surveillance.	Low to Moderate
Safety evidence for ACT exposure in early pregnancy, including observational cohorts and meta-analyses.	Observational safety studies and pooled analyses	Pregnant women with malaria, especially first trimester exposure	No consistent signal of major teratogenic risk from artemisinin-based combinations compared with alternatives; supports prompt effective treatment when benefits outweigh risks.	Moderate

Tafenoquine and primaquine implementation evidence for radical cure of <i>P. vivax</i> .	Trials, pharmacovigilance, operational studies	<i>P. vivax</i> endemic areas	Single-dose tafenoquine improves adherence potential but requires quantitative G6PD testing and is unsuitable in pregnancy and in young children below approved age thresholds.	Moderate
Chemoprevention implementation evidence for SMC and PMC.	Cluster trials, implementation studies, programme evaluations	Children in seasonal and perennial transmission settings	Chemoprevention substantially reduces clinical malaria where drug efficacy is preserved; impact depends on coverage, adherence, timing, and resistance.	High for SMC; Moderate for PMC
Vector-control resistance monitoring studies.	Entomological surveillance	Malaria-endemic countries	Pyrethroid resistance remains widespread, supporting insecticide-resistance management and deployment of next-generation nets where appropriate.	Moderate

2.3 Evidence Gaps and Future Research Needs

Diagnosis

- Better field diagnostics for:
 - Non-*falciparum* species.
 - Low-density asymptomatic infection.
 - Mixed-species infection.
 - G6PD deficiency in peripheral settings.
- More precise thresholds for switching away from HRP2-based rapid diagnostic tests in areas with *pfhrp2/pfhrp3* deletions.
- Operational evidence on integrated fever management where malaria positivity declines.

Treatment

- Continued monitoring of ACT efficacy in Africa, especially in areas with emerging partial artemisinin resistance.
- Evidence on optimal ACT policy when partner-drug resistance and artemisinin partial resistance coexist.
- More data on:

- Triple ACTs.
- New antimalarial combinations.
- Severe malaria adjunctive therapies.
- Treatment in infants below standard dosing thresholds.
- Treatment in the first trimester of pregnancy.
- Better pharmacokinetic data in malnourished children and pregnant women.

***P. vivax* radical cure**

- Safer and more deployable G6PD testing.
- Real-world effectiveness of tafenoquine compared with primaquine.
- Strategies for radical cure in children, pregnant women, and lactating women.
- Relapse-prevention strategies in co-endemic areas with limited laboratory capacity.

Vaccines

- Head-to-head effectiveness data for RTS,S/AS01 versus R21/Matrix-M are limited.
- Longer-term safety and effectiveness data are needed.
- Best schedules for:
 - Perennial transmission.
 - Seasonal transmission.
 - Integration with SMC.
 - Catch-up campaigns.
- Durability of protection and booster timing need further evaluation.
- Impact on severe malaria, mortality, anemia, and health-system burden should be monitored.

Chemoprevention

- Optimal drug regimens in areas with SP or amodiaquine resistance.
- Long-term resistance selection pressure from repeated chemoprevention.
- Best combinations of vaccines plus SMC or PMC.
- Equity and delivery models for hard-to-reach populations.

Vector control

- Comparative effectiveness of different dual-active-ingredient nets.
- Resistance management for chlorfenapyr and other newer insecticides.
- Cost-effectiveness of next-generation nets versus IRS versus combined approaches.
- Evidence for larval source management outside highly specific ecological settings.
- Better tools for outdoor-biting and early-evening-biting vectors.

Surveillance and elimination

- More accurate methods for detecting residual transmission.
- Genomic surveillance integration into national programmes.
- Strategies for imported malaria in elimination settings.
- Evidence-based outbreak thresholds and response packages.

2.4 Updated Recommendations Based on New Evidence

Original WHO-based recommendation	New Evidence Impact	Updated Recommendation	GRADE Evidence Quality	Strength
Use parasitological confirmation before treatment where possible.	No evidence weakens this recommendation; HRP2-deletion evidence emphasizes need for locally appropriate test selection.	Continue universal parasitological confirmation where feasible; adapt RDT choice based on <i>pfhrp2/pfhrp3</i> surveillance.	High for testing; Moderate for HRP2 policy adaptation	Strong
Treat uncomplicated <i>P. falciparum</i> malaria with effective ACT.	Emerging artemisinin partial resistance increases importance of therapeutic efficacy monitoring but does not displace ACTs as first-line therapy where efficacy remains adequate.	Continue ACTs according to national policy; intensify efficacy surveillance and prepare policy change if treatment failure thresholds are exceeded.	High	Strong
Treat severe malaria with parenteral artesunate.	No newer evidence changes the mortality benefit of artesunate over quinine.	Continue immediate parenteral artesunate followed by complete oral ACT.	High	Strong
Use primaquine or tafenoquine for <i>P. vivax</i> radical cure after G6PD assessment.	Tafenoquine improves adherence but requires strict G6PD testing and eligibility screening.	Use tafenoquine only where quantitative G6PD testing and pharmacovigilance are available; otherwise use primaquine regimens appropriate to G6PD status.	Moderate	Strong
Provide IPTp-SP in eligible pregnant women in moderate-to-high transmission African settings.	Evidence continues to support IPTp-SP where SP remains effective. Resistance and HIV cotrimoxazole coadministration remain constraints.	Continue IPTp-SP from the second trimester at each ANC contact at least one month apart, unless contraindicated.	High	Strong

Provide SMC in highly seasonal transmission areas.	Programme data continue to show high impact when coverage and timing are strong.	Continue SMC and consider locally appropriate age expansion where burden extends beyond children under 5 and efficacy is preserved.	High	Strong
Provide PMC in perennial transmission settings where appropriate.	Evidence supports burden reduction, but implementation context matters.	Implement PMC based on local burden, age pattern, delivery feasibility, and drug-resistance profile.	Moderate	Conditional
Use RTS,S/AS01 in children in moderate-to-high transmission settings.	Implementation evidence supports feasibility and public-health impact.	Continue RTS,S/AS01 where selected by national programmes.	Moderate	Strong
Use R21/Matrix-M as an additional malaria vaccine option.	Phase 3 evidence supports efficacy and acceptable short-term safety.	R21/Matrix-M may be used as an alternative WHO-recommended vaccine option according to national policy, supply, cost, and delivery feasibility.	Moderate	Strong
Use insecticide-treated nets as core vector control.	No evidence weakens core role; resistance evidence supports newer net classes.	Continue universal ITN coverage; use PBO or dual-active-ingredient nets in resistance settings based on WHO prequalification and national entomological data.	High for ITNs; Moderate for next-generation nets	Strong for ITNs; Conditional for product selection
Use IRS where epidemiologically and operationally appropriate.	Evidence remains context-specific.	Continue IRS as core or supplementary intervention where coverage, residual activity, vector behavior, and insecticide-resistance management support use.	Moderate	Conditional

Maintain surveillance in all settings.	Resistance, diagnostic-deletion, and vaccine-rollout evidence make surveillance more important.	Strengthen integrated surveillance covering cases, deaths, drug efficacy, molecular resistance, vector resistance, vaccine coverage, and intervention impact.	Moderate	Strong
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Practical WHO-Based Clinical Management Summary

Suspected uncomplicated malaria

1. Assess danger signs.
2. Test with microscopy or quality-assured RDT.
3. If positive:
 - Identify species where possible.
 - Treat according to species and national first-line regimen.
4. If negative but clinical suspicion is high:
 - Reassess diagnosis.
 - Repeat testing if symptoms persist or exposure risk is high.
 - Manage other causes of fever.

Uncomplicated *P. falciparum* malaria

- Use nationally recommended ACT, such as:
 - Artemether-lumefantrine.
 - Artesunate-amodiaquine.
 - Dihydroartemisinin-piperaquine.
 - Artesunate-mefloquine.
 - Artesunate-sulfadoxine-pyrimethamine where still effective.
 - Pyronaridine-artesunate where approved and appropriate.
- Avoid oral artemisinin monotherapy.
- Ensure full adherence to the complete regimen.

Severe malaria

Treat as a medical emergency.

1. Give parenteral artesunate immediately.
2. Manage airway, breathing, circulation, glucose, seizures, anemia, acidosis, renal impairment, and secondary infection.
3. Once oral therapy is possible, complete treatment with a full course of effective ACT.
4. Do not use corticosteroids or other adjunctive therapies without evidence-based indication.

P. vivax* and *P. ovale

1. Treat blood-stage infection.
2. Assess G6PD status.
3. Provide radical cure:
 - Primaquine regimen where appropriate; or
 - Tafenoquine single dose only if eligible and quantitative G6PD normal.
4. Avoid primaquine and tafenoquine in pregnancy.
5. Use specialist or national guidance for lactating women and infants.

Prevention package

- Insecticide-treated nets for at-risk populations.
 - IPTp-SP for eligible pregnant women.
 - SMC in seasonal transmission settings.
 - PMC in selected perennial transmission settings.
 - Vaccination with RTS,S/AS01 or R21/Matrix-M where nationally adopted.
 - Surveillance-guided vector control.
 - Prompt diagnosis and effective treatment.
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Key References

1. World Health Organization. **WHO Guidelines for malaria**. Geneva: World Health Organization; latest verifiable consolidated release used in this report: **13 March 2024**.
2. World Health Organization. **WHO Guidelines for malaria vector control**. Geneva: World Health Organization; 2019.
3. World Health Organization. **Guidelines for the treatment of malaria**. 3rd ed. Geneva: World Health Organization; 2015.
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6. World Health Organization. **WHO recommends R21/Matrix-M vaccine for malaria prevention in updated advice on immunization**. Geneva: World Health Organization; 2023.
7. Datto MS, et al. **Safety and efficacy of malaria vaccine candidate R21/Matrix-M in African children: a multicentre, double-blind, randomised, phase 3 trial**. *The Lancet*. 2024.
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9. World Health Organization. **Global report on antimalarial drug efficacy and drug resistance**. Geneva: World Health Organization; latest available editions.
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